Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
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Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2024-25 (ODD)**	**Course Name**	**Web Technology Lab**
Semester	**3**	**Course Code**	**23CT1301**
Roll No	**63**	**Name of Student**	**Shakul Vilas Rathod**

Practical Number	**4B**
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	**To design an interactive To-Do List web application using HTML, CSS, and Vanilla JavaScript Document Object Model (DOM) manipulation methods for adding, completing, and deleting tasks dynamically.**
Problem Definition	**Develop a browser-based task management interface where users can dynamically enter tasks via an input field. Write core JavaScript logic to manipulate DOM nodes directly (using createElement, appendChild, and remove), handle interactive user events (clicking to strike through completed tasks and deleting elements), and validate against empty inputs without refreshing the page.**
Theory (100 words)	**The Document Object Model (DOM) represents an HTML document as a hierarchical node tree, allowing programming languages like JavaScript to dynamically modify page structure, style, and content. Vanilla JavaScript offers native API methods to interact with this tree without external libraries.** **Node creation and rendering are achieved using document.createElement(), property assignments (textContent, className), and attachment methods like appendChild(). Dynamic interactions rely on event-driven programming: addEventListener() attaches listeners to specific user events (click, keydown). Runtime modifications—such as toggling CSS classes via classList.toggle() for state changes and unmounting nodes using node.remove()—demonstrate asynchronous, client-side UI updates.**

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Procedure and Execution

(100 Words)

HTML Layout: Create a primary container holding an input field, an add button, and an empty unordered list (<ul id="taskList">).

Component Styling: Define standard styling for list items, action buttons, and a .completed class with a line-through text decoration.

Target Nodes: Select the input element, button, and list container in JavaScript via document.getElementById().

Task Injection Logic: Define an addTask handler that validates input text, generates <li>, , and <button> elements, sets their attributes, and mounts them using appendChild().

State & Deletion Handling: Bind click listeners to dynamically created nodes—one to toggle completion status via class manipulation and another to remove the parent <li> element.

HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>To-Do List App</title>
  <link rel="stylesheet" href="style.css" />
</head>
<body>

  <div class="todo-app">
    <h2>My To-Do List</h2>

    <div class="input-container">
      <input type="text" id="taskInput" placeholder="Add a new task..." />
      <button id="addBtn">Add Task</button>
    </div>

    <ul id="taskList"></ul>
  </div>

  <script src="script.js"></script>
</body>
</html>
```

CSS:

```
* {
```

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```
  box-sizing: border-box;
  margin: 0;
  padding: 0;
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
}

body {
  background: #f1f5f9;
  display: flex;
  justify-content: center;
  align-items: center;
  min-height: 100vh;
  padding: 20px;
}

.todo-app {
  background: #ffffff;
  width: 100%;
  max-width: 420px;
  padding: 24px;
  border-radius: 10px;
  box-shadow: 0 8px 24px rgba(0, 0, 0, 0.08);
}

.todo-app h2 {
  color: #1e293b;
  margin-bottom: 20px;
  text-align: center;
}

.input-container {
  display: flex;
  gap: 8px;
  margin-bottom: 20px;
}

.input-container input {
  flex: 1;
  padding: 10px 14px;
  border: 1.5px solid #cbd5e1;
  border-radius: 6px;
  outline: none;
  font-size: 14px;
}
```

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```
.input-container input:focus {
  border-color: #3b82f6;
}

.input-container button {
  padding: 10px 16px;
  background-color: #2563eb;
  color: #ffffff;
  border: none;
  border-radius: 6px;
  cursor: pointer;
  font-weight: 600;
  transition: background 0.2s;
}

.input-container button:hover {
  background-color: #1d4ed8;
}

ul {
  list-style: none;
}

li {
  display: flex;
  justify-content: space-between;
  align-items: center;
  background: #f8fafc;
  padding: 10px 12px;
  border-radius: 6px;
  border: 1px solid #e2e8f0;
  margin-bottom: 8px;
  transition: all 0.2s;
}

li.completed .task-text {
  text-decoration: line-through;
  color: #94a3b8;
}

.task-text {
  flex: 1;
  cursor: pointer;
```

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```
  font-size: 15px;
  color: #334155;
}

.delete-btn {
  background: #ef4444;
  color: #ffffff;
  border: none;
  border-radius: 4px;
  padding: 4px 8px;
  font-size: 12px;
  cursor: pointer;
}

.delete-btn:hover {
  background: #dc2626;
}
```

Javascript:

```
const taskInput = document.getElementById("taskInput");
const addBtn = document.getElementById("addBtn");
const taskList = document.getElementById("taskList");

// Add Task Function
function addTask() {
  const taskText = taskInput.value.trim();

  if (taskText === "") {
    alert("Task cannot be empty!");
    return;
  }

  // 1. Create elements using DOM methods
  const li = document.createElement("li");
  const span = document.createElement("span");
  const deleteBtn = document.createElement("button");

  // 2. Assign attributes and text
  span.className = "task-text";
  span.textContent = taskText;

  deleteBtn.className = "delete-btn";
  deleteBtn.textContent = "Delete";
```

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```
  // 3. Mark as complete (toggle class on click)
  span.addEventListener("click", function () {
    li.classList.toggle("completed");
  });

  // 4. Delete task element
  deleteBtn.addEventListener("click", function () {
    li.remove();
  });

  // 5. Append elements to DOM
  li.appendChild(span);
  li.appendChild(deleteBtn);
  taskList.appendChild(li);

  // Clear input field
  taskInput.value = "";
}

// Event Listeners
addBtn.addEventListener("click", addTask);

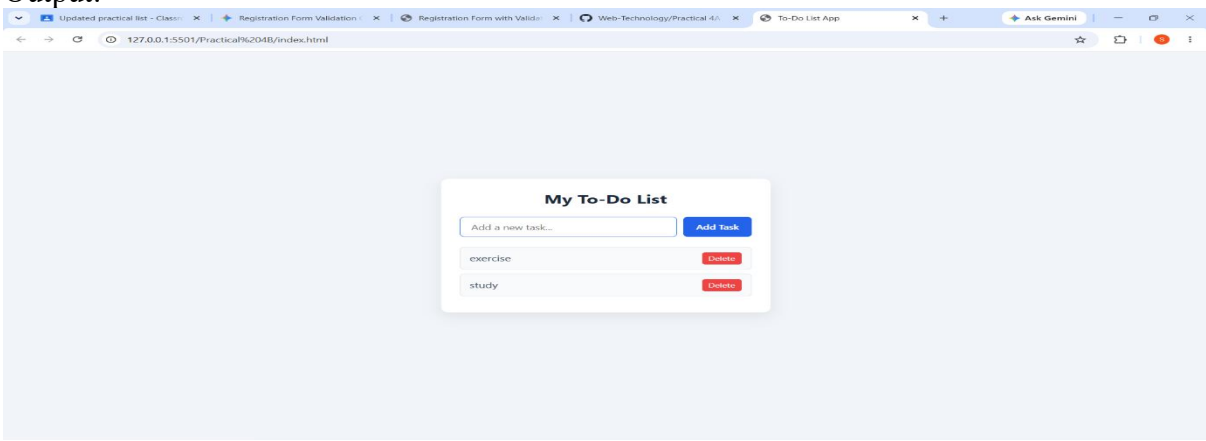
taskInput.addEventListener("keydown", function (e) {
  if (e.key === "Enter") {
    addTask();
  }
});
```

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	Output: 
Output Analysis	**Task Insertion: Entering text and clicking "Add Task" (or pressing Enter) immediately appends a new row containing the task title and a red delete button to the list.** **Blank Input Handling: Submitting an empty field halts execution and shows an alert ("Task cannot be empty!"), preventing blank nodes from being added.** **Completion Toggle: Clicking the task text toggles the .completed class, applying a strikethrough effect and graying out the text. Clicking again reverts the state.** **Task Removal: Clicking the "Delete" button invokes node.remove(), instantly deleting that particular list element from the active DOM tree.**
Link of student Github profile where lab assignment has been uploaded	**https://github.com/shakulvilasrathod-art/Web-technology**

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Conclusion	**The interactive To-Do List application was successfully implemented using Vanilla JavaScript DOM manipulation. The experiment verified how dynamic node generation, class toggling, and event delegation facilitate real-time UI updates and responsive client-side interactivity without server round-trips.**
Plag Report (Similarity index < 12%)	
Date	**21-09-2026**